Design Detailed Documentation - ROBOCON 2025

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ABU ROBOCON 2025 introduces the exciting theme of "Robot Basketball", where two teams compete against each other, each team consists of two robots namely Robot 1 and Robot 2. Robots compete to outscore the other by shooting a ball into the opponent's basket while defending their own. Robot 1 focusing on dribbling, passing, shooting and defending. Robot 2 performs skills like jumping and dunking. Robots must coordinate together to maximize the score, all while adhering to the rules.

I. Design of Robot 1:

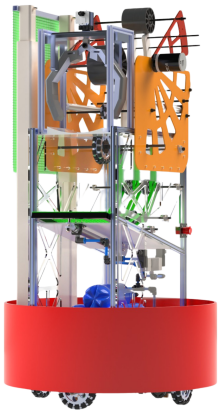


Figure 1: Robot-1

- **Overall dimensions: 800 mm (diameter) × 1440 mm (height).**
- **Extended dimensions: 1070 mm (diameter) × 2350 mm (height).**
- **Estimated weight: 28 Kg.**

1. Type of Drive:

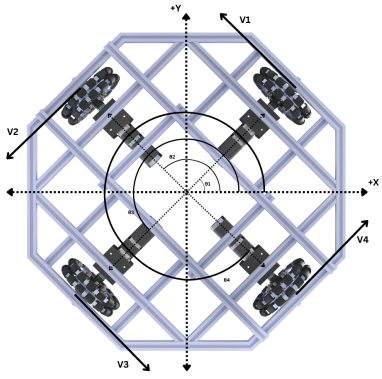


Figure 2: Four Wheel Omni-directional Drive

Robot 1 uses **Four Wheel Omni-directional Drive** mounted on a custom-made chassis. The calculations for the four-wheel Omni drive are as follows:

$$\vec{v} = \vec{\omega} \times \vec{R}$$

Here, $\vec{R}$ represents the distance vector between the center of the base and the Omni wheels of the robot. $\vec{V}$ can be resolved in the X and Y directions.

X-component: $v_x + \omega r \sin\theta_i r = v_i \cos\theta_i - v_i r \sin\theta_i$

Y-component: $v_y + \omega r \sin\theta_i r = v_i \cos\theta_i + v_i r \sin\theta_i$

$$v_i = v_x \cos\theta_i + v_y \sin\theta_i + \omega r$$

where,

v_i : Linear velocity of the ith*wheel*.

v_x : Linear velocity of the robot along the X-axis.

v_y : Linear velocity of the robot along the Y-axis.

ω : Angular velocity of the robot about the Z-axis.

r : Radius of the wheel.

θ_i : Angle of ith wheel relative to X-axis of the robot.

$$V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} \cos\theta_1 & \sin\theta_1 & r \\ \cos\theta_2 & \sin\theta_2 & r \\ \cos\theta_3 & \sin\theta_3 & r \\ \cos\theta_4 & \sin\theta_4 & r \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix}$$

2. Actuators:

- **Planetary Motors:** Four mini Planetary motors of **750 RPM** are used to drive Robot 1 using inbuilt encoder's (**325 PPR**) RPM feedback.
- **DC Geared motor:** Two motors of **1000 RPM** and **0.7 kg-cm torque**, are used for cascade and feeder system.
- **Stepper Motor:** Stepper motors of **4.2 kg-cm torque** are used to change the angle of arc shooter.
- **6375 BLDC Motor: 4500 rpm and 4.9 N-m torque** for shooting.
- **Pneumatic Cylinder:** One compact pneumatic cylinder **(Bore-diameter (BD): 12mm, Stroke length (SL): 50mm)** and three pencil pneumatic cylinder **(Bore-diameter (BD): 25mm, Stroke length (SL): 300mm)**

3. Sensors:

- **IMU**: **BNO055** and **ADXRS453Z** are used for robot localization.
- **Rotary Encoder**: **Two** Rotary Encoders of **2500 PPR** are mounted perpendicular to each other at the robot's bottom for odometry.
- **Processor**: **Raspberry Pi 4 Model B, Jetson Nano**, and **Intel NUC** handles object detection and **ROS implementation** in the robot.
- **Controllers**: **Portenta H7** handles real-time movement, **ESP32** manages wireless communication and sensors, while **Arduino Mega**, **UNO** and **Due** handle general control.

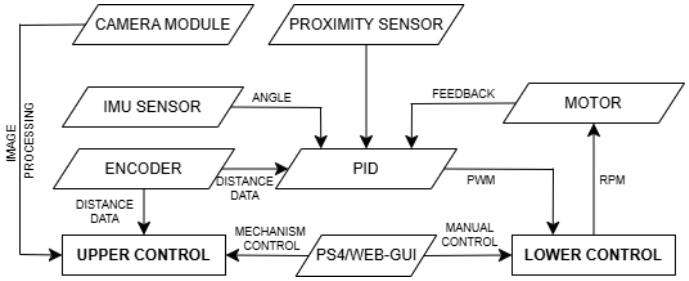


Figure 3: Flowchart for Robot 1 and Robot 2

4.Communication & Algorithms:

- **Stereo camera** converts video into frames using **open CV** while storing the images as dataset used for model training.
- Initiate state vector variables and using **Projectile motion** with gravity, Sequence of 10 Kalman-filtered states (**100 ms history**) for 20 future time steps.
- Estimate the **landing zone coordinates**, adjust wheel speeds using **Proportional-Integral-Derivative algorithm control** to track the target, and open the net to catch the ball.
- **CAN communication** protocol's built-in error detection, priority messaging, and fault handling ensure reliable communication, making it ideal for **ESP32** and **Arduino** due-based omni drive control.

5. Dribbling Mechanism:

5.1 Overview

This mechanism consists of two frames, base plate, two-claw gripper, T-slot sliders, receiving net and various pneumatic actuators.

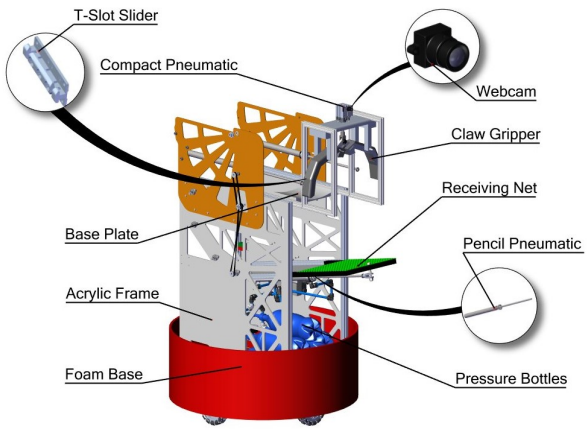


Figure 4: Dribbling Mechanism

5.2 Mechanism

It consists of **two-claw 3D printed gripper**, **aluminium composite base plate (260mm x 160mm)** and two aluminium box section frames with T-slot sliders driven by two pencil pneumatic cylinder **(BD:25mm, SL:300mm).** A compact pneumatic cylinder **(BD:12mm, SL:50mm)** positioned above the gripper, another pencil pneumatic **(BD:25mm, SL:300mm)** drives the net mechanism positioned at **720mm**.

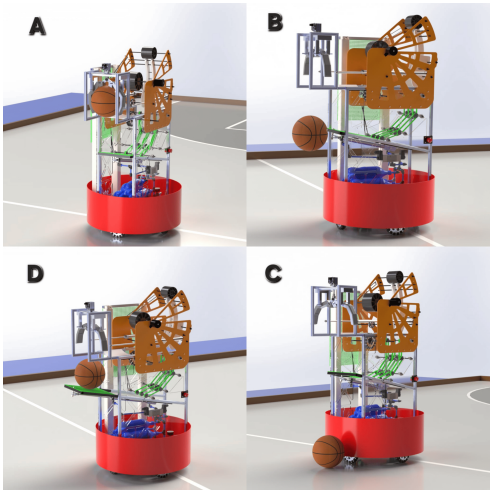


Figure 5: Dribbling and ball handling Mechanism

5.3 Task Objective

The dribbling operation, necessary for shooting, begins with the ball placed onto base plate at a height of **1000 mm**. **T-slot slider** enable translatory motion of the gripper. A compact pneumatic cylinder applies **downward force** through the **gripper links**, ensuring precise dribbling. Upon rebound, a pencil pneumatic cylinder activates the **net** to capture the ball. The ball is then transferred to the shooting mechanism through a **feeder drive system**.

5.4 Justification

The **two claw gripper** provides a good contact area for gripping, while the T-slot sliders enable **translatory motion**. Pneumatic actuators are precise and fast. This mechanism also provides sufficient force, allowing the ball to dribble more than **700 mm** as per given rules. The following is the equation for the same:

$$u = \sqrt{2g\left(\frac{h_2}{e^2} - h_1\right)}$$

where **u** is initial velocity required for dribbling, h_2 is height obtained by the ball after dribbling, h_1 is height of ball before dribbling (i.e., $h_1 = 1160mm$) and e is coefficient of restitution.

6. Shooting Mechanism:

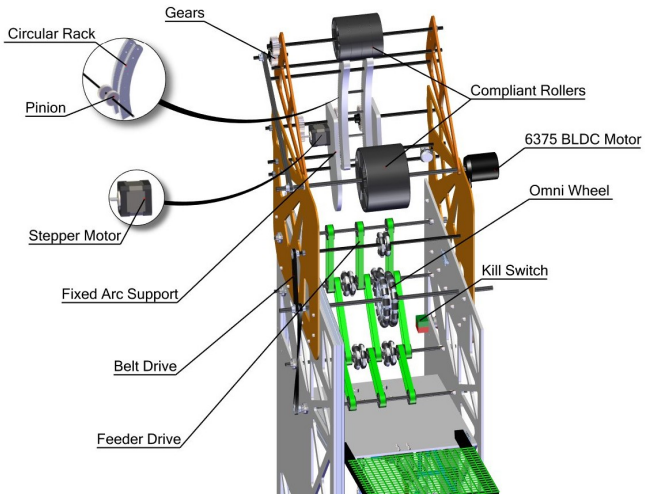


Figure 6: Shooting Mechanism

6.1 Overview

The shooting mechanism consists of a BLDC motor, compliant rollers, a circular rack and pinion and a feeder drive system with omni wheels.

6.2 Mechanism

The mechanism uses a 6375 BLDC motor **(4500 RPM, 4.9 N-m torque)** to drive **EPDM DURO 60 (Ethylene Propylene Diene Monomer Durometer 60)** compliant rollers which grip and propel the ball. A circular rack and pinion system **(gear ratio 12.12, module 1.75mm)** adjusts the projectile angle for accurate shot placement.

6.3 Task Objective

After dribbling, the ball is fed into the shooting mechanism via **feeder system**, with the motor driving the ball forward using a **compliant rollers**. The robot orientation will align with the basket and a trajectory equation ensures that the basket's coordinates meet the required conditions. The shooting angle is adjusted through a circular rack and pinion system, with precise control provided by an **integrated stepper motor.**

6.4 Justification

The arc shooter mechanism is accurate, with a strong frame and smooth energy transfer using compliant rollers. The circular rack and pinion system adjusts the angle easily, ensuring reliable and adaptable performance.The following is the equation:

$$y = H + x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

where **y** is vertical distance of ball from ground, **h** is height of Robot 1, **x** is horizontal distance between ball and robot, **u** is initial velocity of ball and θ is angle of projection.

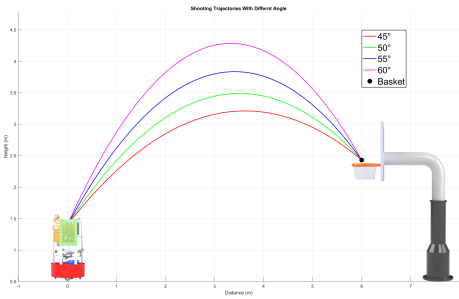


Figure 7: H v/s x graph at different angle of projection

7. Ball Handling:

At the start of the offensive play, the ball will be placed on the **base plate (260mm x 160mm)**, making it easy for the gripper to hold. When the dribbled ball touches the top of the pneumatically driven net, the net will close, guiding the ball to the **feeder drive**. The roller will then feed the ball to the shooting mechanism for either shooting or passing.

II. Design of Robot 2:

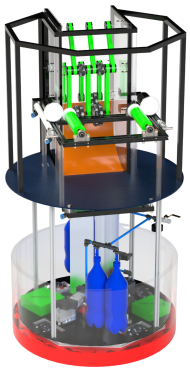


Figure 8: Robot-2

- **Overall dimensions: 800 mm (diameter) × 1470 mm (height).**
- **Extended dimensions: 1150 mm (diameter) × 2200 mm (height).**
- **Estimated weight : 21 Kg.**

1. Type of Drive:

Robot 2 uses **Four Wheel Omni-directional Drive** mounted on a custom-made chassis and the **calculations** for the four-wheel omni drive are same as Robot 1.This setup provides very little chance of toppling, even during fast movements or sudden stops. For Robot 2, which has a **jumping mechanism**, the four-wheel drive ensures stability after landing preventing imbalance. Additionally, using the same drive system in both robots **simplifies motion control and programming**, making it easier to synchronize their movements and optimize performance.

2. Actuators:

- **Planetary Motors:** Four mini Planetary motors of **750 RPM** are used to drive Robot 2 using inbuilt encoder's (**325 PPR**) velocity feedback. One for dunking mechanism.
- **Gear motor:** Two gear motors of **1000 RPM** and **0.7 kg-cm torque**, are used for dribbling.
- **Servo Motor:** Two servo motors of **25 kgf-cm torque** are used in rachet mechanism.
- **T**wo 6375 BLDC Motor: 4500 rpm and 4.9 N-m torque is used to lift the piston.
- **Pneumatic Cylinder:** One compact pneumatic cylinder **(Bore-diameter (BD): 12mm, Stroke length (SL): 50mm)** and three pencil pneumatic **(Bore-diameter (BD): 25mm, Stroke length (SL): 300mm)**

3. Sensors :

- **IMU**: The robot uses a **BNO055 and an ADXRS453Z** to accurately track its position and movement.
- **Rotary Encoder**: **Two** rotary encoders **(325 PPR)** are mounted perpendicular to each other at the bottom of the robot to measure movement and direction.
- **Processor**: The robot's main processors include **Raspberry Pi 4 Model B, Jetson Nano and Intel NUC.**These are used for running **object detection algorithms and implementing ROS (Robot Operating System).**
- **IR and Proximity Sensor** used for basket detection.
- **Controllers**: **Portenta H7** (ideal for real-time processing in movements), **ESP32** (manages wireless communication through Websocket and sensor integration), **STM32H7** (used to generate precise PWM signals for lower control), **Arduino Mega, Arduino UNO, Arduino Due.**

4. Jumping Mechanism:

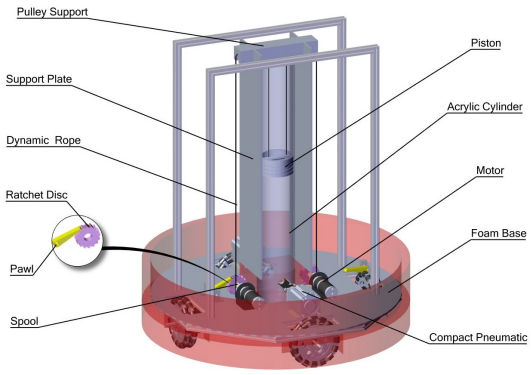


Figure 9: Jumping Mechanism.

4.1 Overview

The jumping mechanism consists of a **700 mm** long hollow cylinder, a motor-driven spool, **dynamic rope drive** attached to piston, a ratchet mechanism, and a vacuum system, to lift a **21 kg** robot.

4.2 Mechanism

The 700 mm long hollow cylinder is made up of **acrylic**. The BLDC motor-driven spool **(diameter: 20mm)** two dynamic ropes.The ratchet system is mounted on the chassis. This mechanism uses reverse pneumatic system,relying on pressure difference between atmospheric pressure and reduced pressure inside.

4.3 Task Objective

The mechanism uses a reverse pneumatic system, relying on the pressure difference between atmospheric pressure and the reduced pressure inside. A motor-driven spool wind the ropes attached to the piston, creating a vacuum. A ratchet mechanism locks the spool to maintain tension thus allowing atmospheric pressure to lift the chassis efficiently.

4.4 Justification

Reverse pneumatics offers energy-efficient actuation by using atmospheric pressure for power, eliminating external actuators. A motor pulley ratchet system prevents backsliding, enabling stable, controlled jumping. The tension equation is

$$T = \frac{(m - M)g + \Delta PA}{2}$$

where, m is piston mass, M is counter mass, ΔP is pressure difference, and A is piston area, calculated as

$$\Delta P = P_0 \times \left[1 - \left(\frac{h_i}{h_f}\right)^{\gamma}\right]$$

where P_o is atmospheric pressure, h_i and h_f are initial and final piston heights, and γ (1.4 for air) is the adiabatic constant.

5. Dribbling Mechanism:

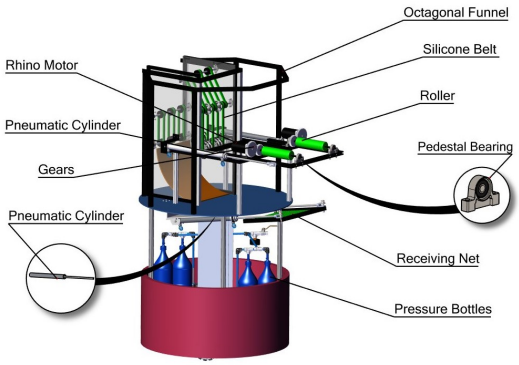


Figure 10: Dribbling Mechanism

5.1 Overview

The mechanism consists of a roller frame, pneumatic actuators, T-slot sliders, and a receiving net.

5.2 Mechanism

A silicone roller forces the ball downwards, with T-slot sliders enabling translatory motion. A pencil pneumatic cylinder **(BD:25mm and SL:300mm)** opens the net mechanism, positioned at 800 mm to capture the ball.

5.3 Task Objective

The ball is received from Robot 1 and delivered to double roller driven by a gear motor, which dribbles the ball. At a height of 800mm, where recieving net controlled by pnuematic cylinder catches the ball and feeds it into the dunking system.

5.4 Justification

This mechanism efficiently performs two tasks **(receiving and dribbling)** simultaneously. The rebound catch, facilitated by a pneumatically controlled net, is both rapid and precise.

6. Ball Handling:

In Robot 2, ball is first received in an octagonal shaped funnel, then delivered to the double-roller frame system, which has already moved forward from its initial position and actively dribbling. Once dribbled, the receiving net collects the ball and feeds it into the belt system for dunking.

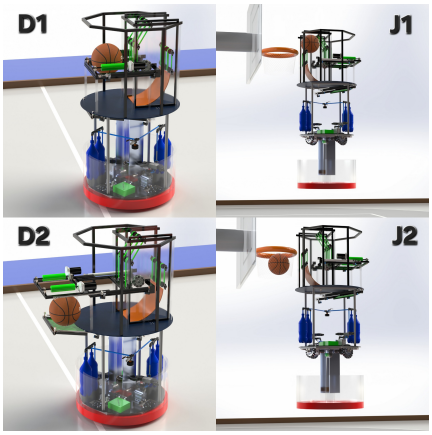


Figure 11: D:Dribbling, J:Jumping

7. Dunking:

After a successful dribble, the jumping mechanism is initiated simultaneously loading the ball to the belt drive. The robot is lifted by the jumping mechanism. Once it reaches the optimal position, the belt drive engages and finishes the dunk.

8. Ball Passing (R1 to R2):

Robot 1 **continuously tracks** the ball and shares it's positions while maintaining a **1000 mm** distance, **enabling real-time position data exchange**. Maintaining a minimum distance of **1000 mm**, robot 1 adjusts its orientation and uses the shooting mechanism to launch the ball towards robot 2's position. Robot 2 receives the ball through its **octagonal**-shape receiving mechanism.

9. Justification of not including Certain Mechanisms:

Robot 1 is designed for defense with an extended frame and a cascade net lift, making it less suitable for jumping. Therefore, Robot 2 performs the jumping actions.

Robot 2 is designed exclusively for dunking as part of our game strategy, with no shooting mechanism included to minimize weight and maximize efficiency.

Passing: Robot 2 focuses entirely on jumping and dunking, adding a passing mechanism would reduce its performance, making it less effective in executing high-powered dunks.

10. Protection:

Both Robot 1 and Robot 2 have a 25 mm thick foamed polyethylene swimming noodle around their cylindrical frames for impact protection. This lightweight layer absorbs shocks, preventing damage while maintaining stability and mobility.